

JULY HACKATHON 2026 · TRACK A – CRISIS TECH · SOLO
ENTRY

বর্ণমালা · Bornomala

An offline Bangla SMS app with three jobs: compress ordinary messages so
5× more fits in one text, compose emergency reports any phone can read,
and let one volunteer relay many households' reports in a single SMS.

Everything runs on the phone. No server, no internet, no account, no
message ever leaves the device except through your own SMS app.

mayamoho.github.io/bornomala · github.com/Mayamoho/bornomala

In July 2024 the state cut the mobile internet.

Broadband went dark too. What kept working was the oldest part of the phone network: voice calls and text messages. People fell back to texting.

Every crisis tool assumes it can build a new network. In July, nobody could.

And it costs twice as much to text in Bangla.

Text messages were built around an old character set that only covers English. Write one Bangla letter and the whole message switches to a heavier format:

160

characters per segment, English

70

characters per segment, Bangla

2.3×

more expensive, per message

More segments means more cost, more retries, and more failures on a congested network — exactly when SMS is the only channel left.

A mesh reaches the room. A text reaches your mother.

Half of crisis tech is offline mesh messengers — Bluetooth or Wi-Fi passing messages phone to phone. Good idea, three conditions attached:

	BLUETOOTH MESH APP	BORNOMALA
Who needs the app	both people, installed <i>before</i> the disaster	only the sender
How far it reaches	about 100 metres	the whole country
Who it can reach	strangers nearby who also installed it	any phone that can receive a text
Can it call 999	no	yes, one tap

So Bornomala does not build a network. It uses the one that already reaches everybody — text messages and voice calls — and makes it cheaper.

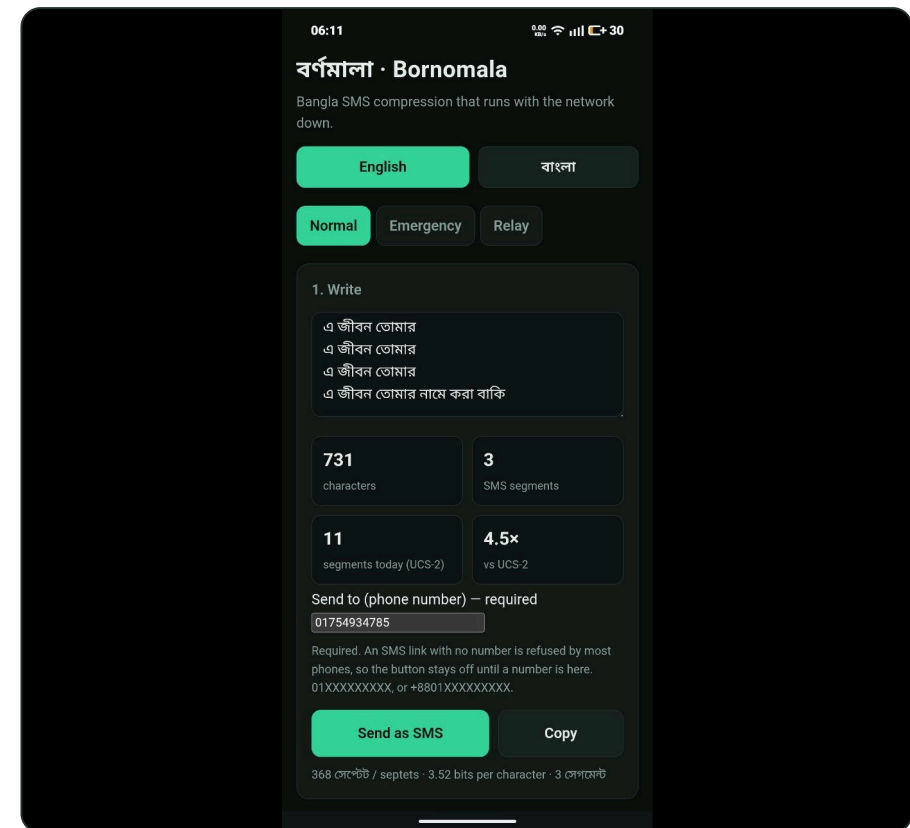
It guesses the next letter, so the letter costs less.

- 1 A small Bangla model rides inside the app. No internet, no server.
- 2 It predicts each next letter. A good guess costs almost nothing to record.
- 3 The bits go out as characters SMS treats as cheap.

আমরা পাঁচজন নিরাপদ আছি, ঢাকা

B*?TΞ9(9Id'0b

731 characters → 3 texts, not 11

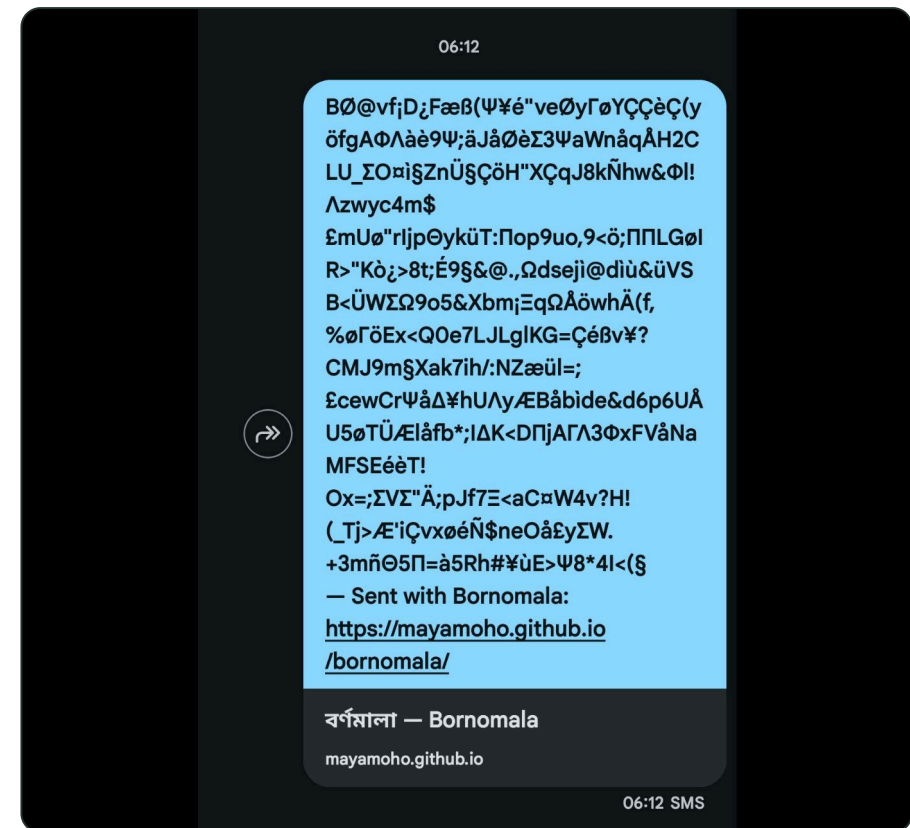


731 Bangla characters: 3 texts, not 11.

Type Bangla. Send it. Read it back.

- **Send** opens your own SMS app, number and message filled in.
- It stays off until a real number is there — a text link with no number silently fails.
- ← Paste the whole message you received. It finds the code, ignores the signature and the map link.
- Anything shared into the app from elsewhere lands in the decode box.

Try it: 01315267528

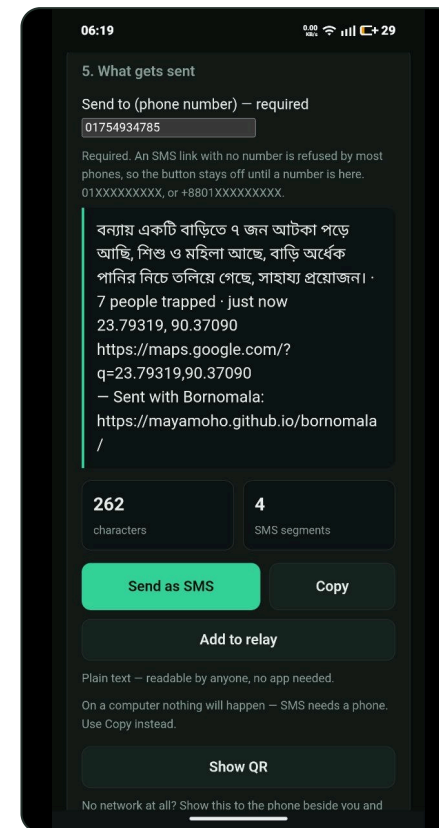


How it lands on the other phone.

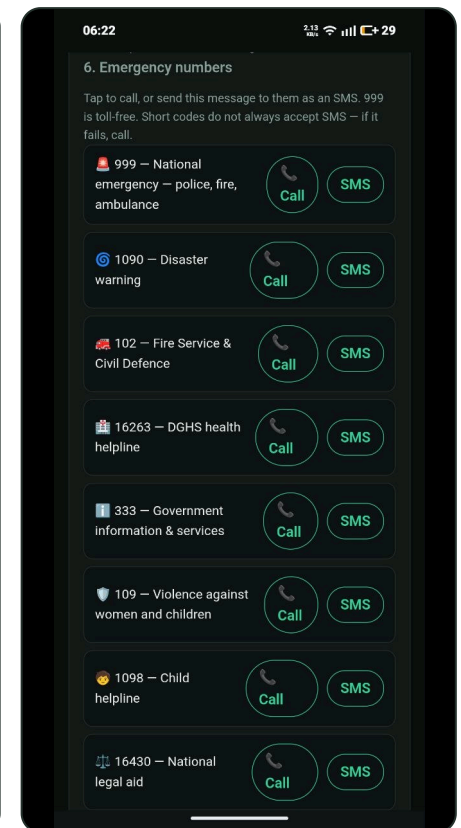
A survivor has no app. So this tab sends words.

- 📍 What is happening, in your own words.
- ✅ One of 32 ready sentences, with a blank to fill.
- 📍 A district, or your live location.
- 🕒 How long ago.
- ☎️ 999 and eight more, one tap to call. **Calls need no internet.**

It refuses to send until those are answered — a report with no place is not a report.



What actually gets sent — plain words.



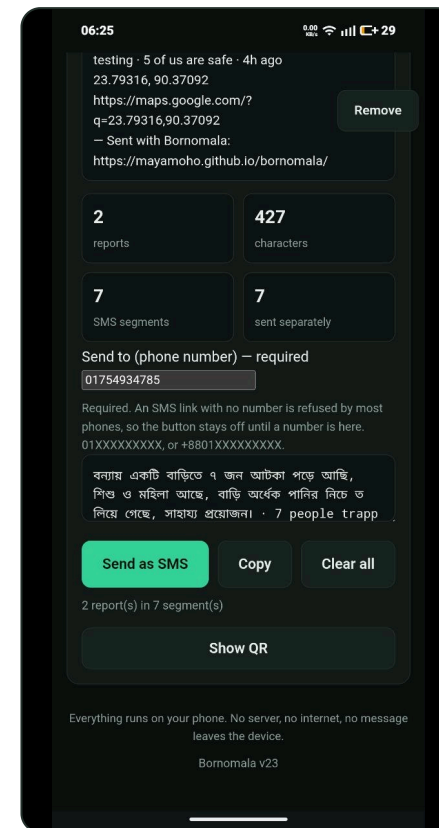
Nine national numbers.

Forty families. One message.

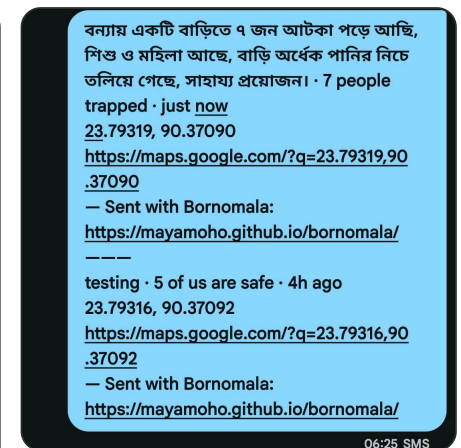
- + Each household is added with one tap from the Emergency tab.
- = They go out together, with one signature for the batch.
- △ A counter shows what sending them one by one would have cost.

Two households → one message

When the towers are jammed, one message that gets through beats forty in a queue.



The queue, and what it costs.



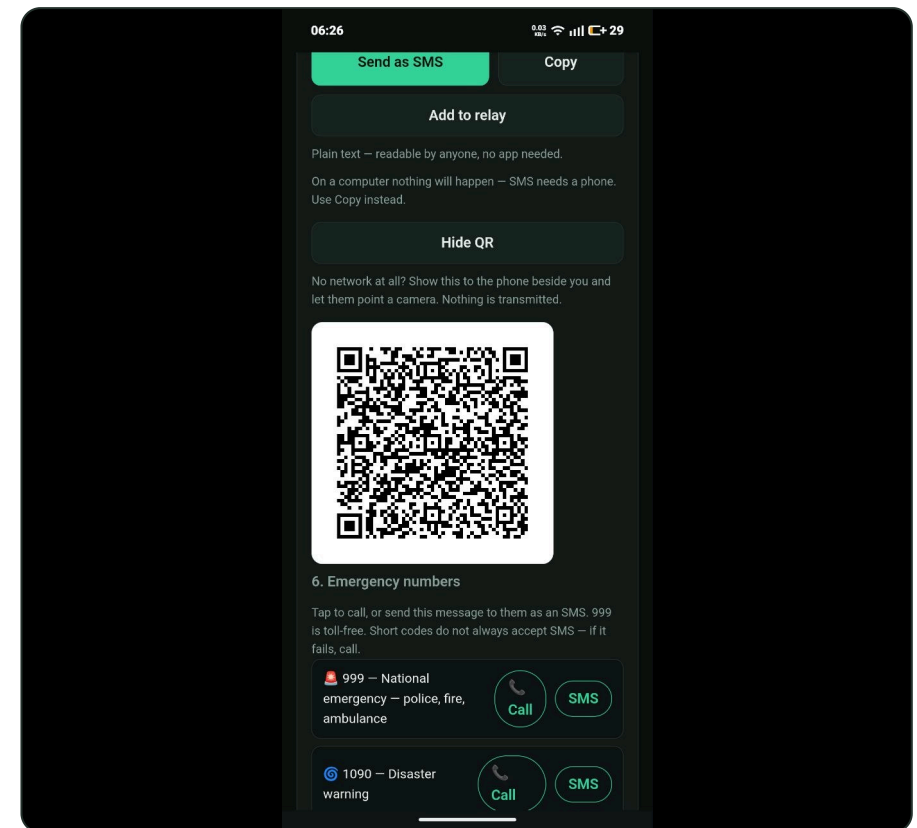
Both arriving as one SMS.

5,000 messages the model had never seen.

	BANGLA IN ONE TEXT	FITS IN ONE
Bornomala	361	100.0%
Every phone today	70	98.2%
gzip	53	99.5%

5.15× more Bangla per text

Ordinary compression does worse than doing nothing — it has no idea what Bangla looks like, and a message this short never repays the overhead.



No network at all? The message becomes a QR.

Each layer below has a floor.

No internet

install it once and it opens
forever after, with no
connection

No mobile data

texts and calls still go, over
2G

No GPS

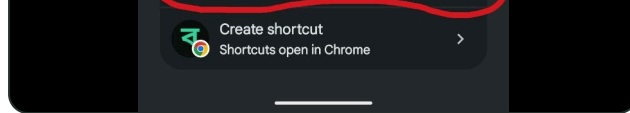
pick your district instead of
using live location

No network at all

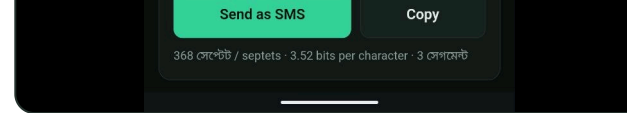
show the QR code and let
the phone beside you
photograph it

Your location always travels twice — as a map link *and* as plain numbers beside it — because the numbers still make sense on a phone that cannot open the link.

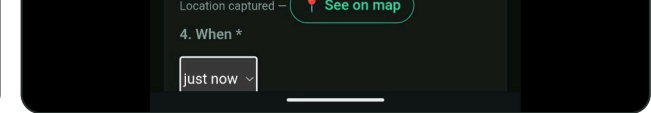
No accounts, no tracking, no ads, nothing sent to us. The app is 92 kB and the language model is a separate 1.7 MB, so the Emergency and Relay tabs work before it finishes loading.



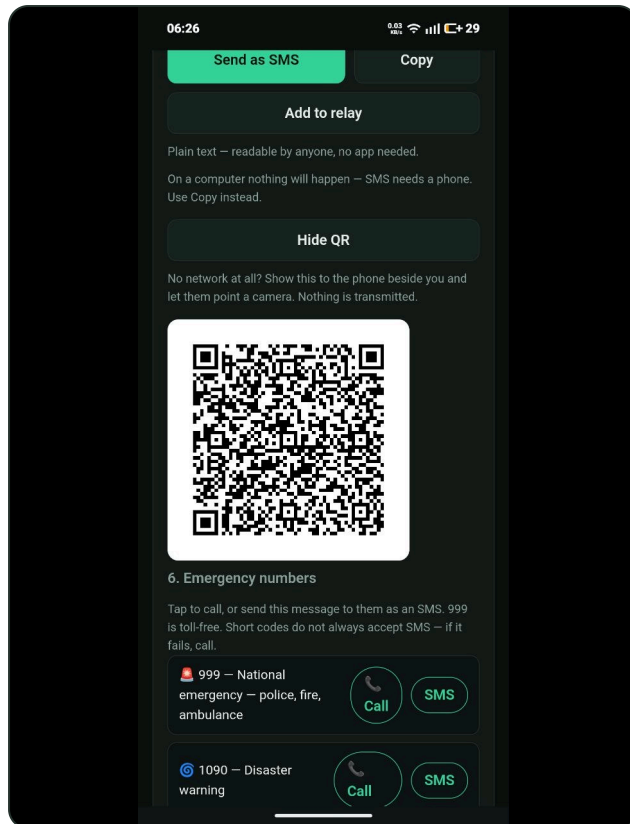
Install from the browser menu — no Play Store, no APK.



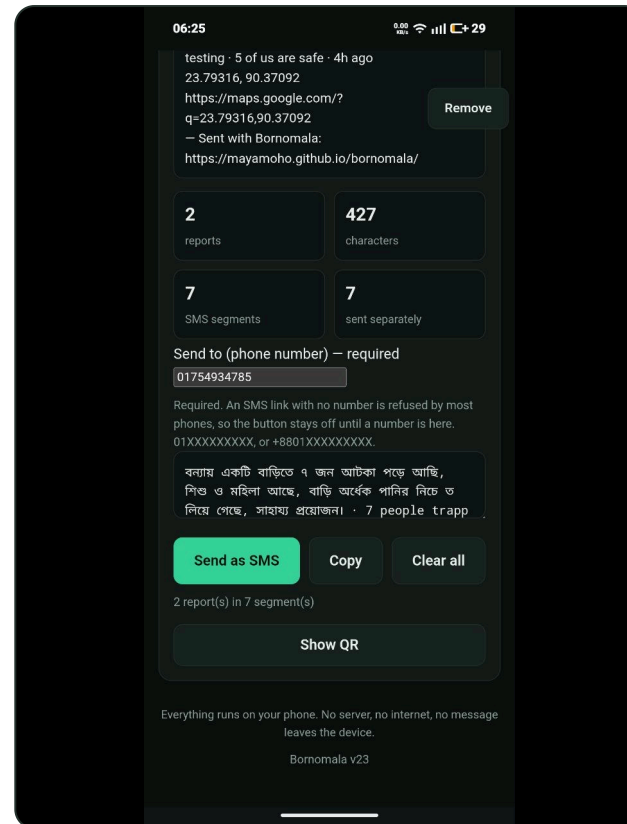
731 Bangla characters: 3 segments instead of 11.



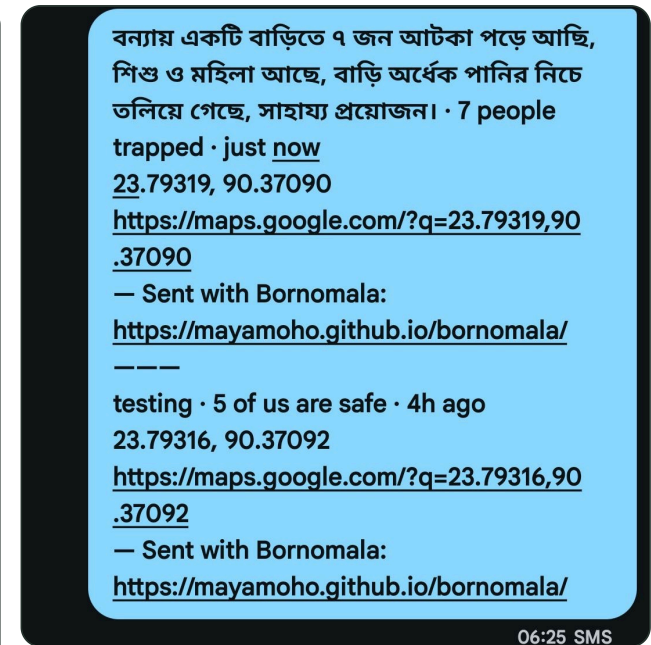
An emergency report, built and previewed.



QR handoff, and the hotlines underneath.



Two households, queued.



The same batch, as it arrives.

